

Slope comp scratchpad

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$$\alpha = \frac{m_{cmp} - m_d}{m_{cmp} + m_c}$$

$$\beta = \frac{m_c + m_d}{m_c + m_{cmp}}$$

$$G_{pk} = \frac{1 - \alpha}{1 + \alpha}$$

$$G_{pk} = \frac{1}{2D \left(\frac{m_{cmp}}{m_d} - 1 \right) + 1}$$

$$m_{cmp}(D) = \frac{1}{D} \left(\frac{1 - G_{pk}}{2G_{pk}} m_d \right) + m_d$$

$$I_{cmp}(D) = T_{SW} \int_0^D m_{cmp}(D) dD$$

$$I_{cmp}(D) = m_d T_{SW} \left(\frac{1 - G_{pk}}{2G_{pk}} \ln(D) + D + \frac{1 - G_{pk}}{2G_{pk}} + 2 \ln \left(\frac{G_{pk} - 1}{2G_{pk}} \right) \right)$$